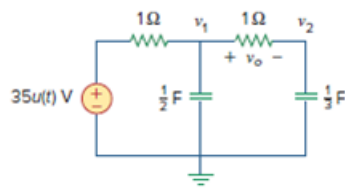


Problem

For $t > 0$, obtain $v_o(t)$ in the circuit of Fig. 8.32. (*Hint*: First find v_1 and v_2 .)

Figure 8.32:



Step-by-step solution

Step 1 of 17

Refer to Figure 8.32 in the text book for the actual circuit.

For $t < 0$, the step function, $u(t)$ is zero that is the voltage source $35u(t)$ is short circuited.

Hence the circuit becomes source free circuit.

Therefore, the initial values are,

$$v_1(0^-) = 0 \text{ V}$$

$$v_2(0^-) = 0 \text{ V}$$

At $t < 0$, the capacitor acts as open circuit. The value of the voltage source is zero since

$u(t) = 0$ for $t < 0$. That is the voltage source acts as short circuit $t < 0$.

Step 2 of 17

At $t = \infty$, as steady state is reached, capacitors are open circuited.

Draw the circuit diagram at $t = \infty$.

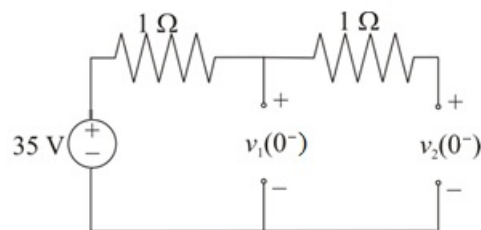


Figure 1

Calculate the steady state value of the voltage.

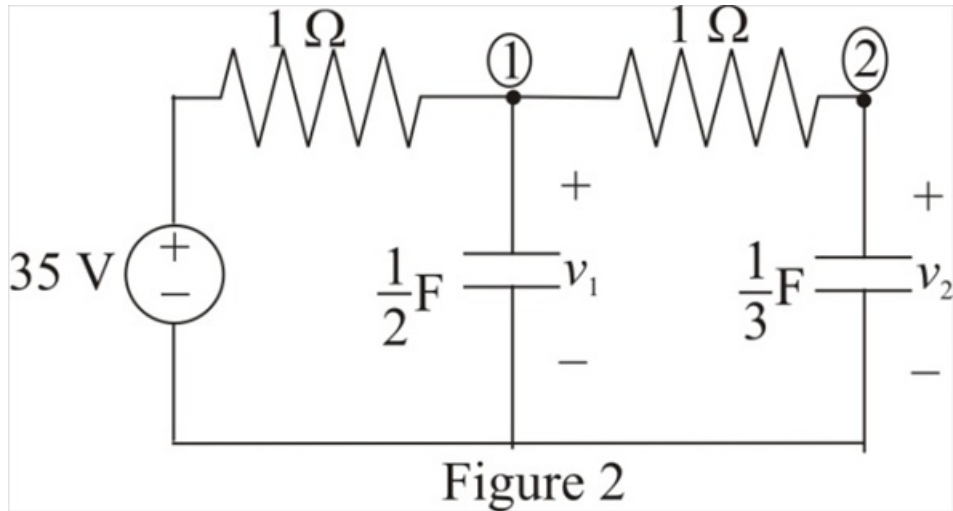
$$\begin{aligned} v_{ss}(t) &= v(\infty) \\ &= 35 \text{ V} \end{aligned}$$

The values of voltages $v_1(\infty)$ and $v_2(\infty)$ are 35 V.

Step 3 of 17

The value of the voltage source is 35 V since $u(t)=1$ for $t \geq 0$.

Draw the circuit diagram at $t > 0$.



Step 4 of 17

The voltage across the capacitor doesn't change instantaneously. Hence, at $t > 0$ the value of the voltages $v_1(0^+)$ and $v_2(0^+)$ are 0 V.

Notice that the voltage across the capacitor is continuous. Therefore, the value of voltages $v_1(0)$ and $v_2(0)$ at $t = 0$ are 0 V.

Step 5 of 17

Apply Kirchhoff's current law at node-1.

$$\frac{v_1 - 35}{1} + \left(\frac{1}{2}\right) \frac{dv_1}{dt} + \frac{v_1 - v_2}{1} = 0$$

$$v_1 - 35 + \left(\frac{1}{2}\right) \frac{dv_1}{dt} + v_1 - v_2 = 0$$

$$2v_1 - v_2 + \left(\frac{1}{2}\right) \frac{dv_1}{dt} - 35 = 0$$

$$\left(\frac{1}{2}\right) \frac{dv_1}{dt} = -2v_1 + v_2 + 35 \dots\dots (1)$$

Calculate the value of $\frac{dv_1}{dt}$ at $t = 0$.

$$\left(\frac{1}{2}\right) \frac{dv_1(0)}{dt} = -2v_1(0) + v_2(0) + 35$$

Substitute 0 V for $v_1(0)$ and $v_2(0)$.

$$\left(\frac{1}{2}\right) \frac{dv_1(0)}{dt} = -2(0) + (0) + 35$$

$$\left(\frac{1}{2}\right) \frac{dv_1(0)}{dt} = 35$$

$$\frac{dv_1(0)}{dt} = 70 \text{ V/s}$$

Therefore, the value of the $\frac{dv_1(0)}{dt}$ is 70 V/s.

Step 6 of 17

Short the voltage source and redraw the circuit to estimate the transient behavior of the circuit.

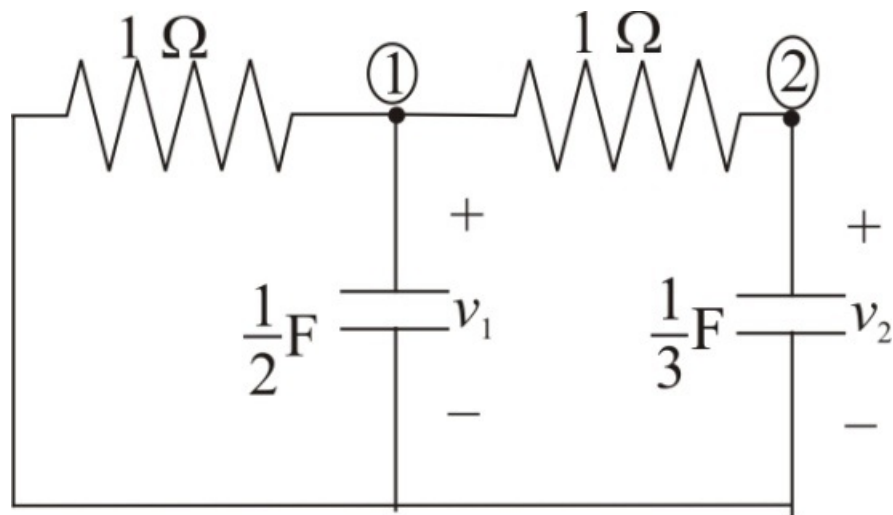


Figure 3

Step 7 of 17

Apply Kirchhoff's current law at node-1.

$$\frac{v_1}{1} + \left(\frac{1}{2}\right) \frac{dv_1}{dt} + \frac{v_1 - v_2}{1} = 0$$

$$v_1 + \left(\frac{1}{2}\right) \frac{dv_1}{dt} + v_1 - v_2 = 0$$

$$2v_1 - v_2 + \left(\frac{1}{2}\right) \frac{dv_1}{dt} = 0$$

$$v_2 = 2v_1 + \left(\frac{1}{2}\right) \frac{dv_1}{dt}$$

Therefore, the expression for the voltage v_2 is $2v_1 + \left(\frac{1}{2}\right) \frac{dv_1}{dt}$.

Step 8 of 17

Apply Kirchhoff's current law at node-2.

$$\left(\frac{1}{3}\right)\frac{dv_2}{dt} + \frac{v_2 - v_1}{1} = 0$$

$$\left(\frac{1}{3}\right)\frac{dv_2}{dt} + v_2 - v_1 = 0$$

Substitute $2v_1 + \left(\frac{1}{2}\right)\frac{dv_1}{dt}$ for v_2 .

$$\left(\frac{1}{3}\right)\frac{d}{dt}\left\{2v_1 + \left(\frac{1}{2}\right)\frac{dv_1}{dt}\right\} + 2v_1 + \left(\frac{1}{2}\right)\frac{dv_1}{dt} - v_1 = 0$$

$$\left(\frac{1}{3}\right)\left\{2\frac{dv_1}{dt} + \left(\frac{1}{2}\right)\frac{d^2v_1}{dt^2}\right\} + v_1 + \left(\frac{1}{2}\right)\frac{dv_1}{dt} = 0$$

$$\frac{2}{3}\frac{dv_1}{dt} + \frac{1}{6}\frac{d^2v_1}{dt^2} + v_1 + \left(\frac{1}{2}\right)\frac{dv_1}{dt} = 0$$

$$\frac{1}{6}\frac{d^2v_1}{dt^2} + \frac{7}{6}\frac{dv_1}{dt} + v_1 = 0$$

$$\frac{d^2v_1}{dt^2} + 7\frac{dv_1}{dt} + 6v_1 = 0$$

Assume $\frac{dv_1}{dt}$ as s .

$$s^2v_1 + 7sv_1 + 6v_1 = 0$$

$$(s^2 + 7s + 6)v_1 = 0$$

Step 9 of 17

Consider the characteristic equation of the transient circuit.

$$s^2 + 7s + 6 = 0$$

$$(s+1)(s+6) = 0$$

The roots of the characteristic equation are $s_1 = -1$ and $s_2 = -6$.

Step 10 of 17

Notice that the roots are negative and real, hence the circuit exhibits over damped response.

Consider the expression for the voltage v_{lt} in transient state.

$$v_{lt} = A_1e^{s_1t} + A_2e^{s_2t}$$

Substitute -1 for s_1 and -6 for s_2 .

$$v_{lt} = A_1e^{-t} + A_2e^{-6t}$$

Step 11 of 17

Consider the expression for the voltage $v_1(t)$ for complete response.

$$v_1(t) = v_{1t}(t) + v_{1ss}(t)$$

Here $v_{1t}(t)$ represents the transient response,

$v_{1ss}(t)$ represents the steady state response that is $v_1(\infty)$

$$v_1(t) = v_1 + v_1(\infty)$$

Substitute 35 V for $v_1(\infty)$ and $A_1 e^{-t} + A_2 e^{-6t}$ for v_{1t} .

$$v_1(t) = A_1 e^{-t} + A_2 e^{-6t} + 35 \dots\dots (2)$$

Step 12 of 17

Recall equation (2).

$$v_1(t) = A_1 e^{-t} + A_2 e^{-6t} + 35$$

Substitute 0 for t .

$$v_1(0) = A_1 e^{-t(0)} + A_2 e^{-6(0)} + 35$$

Substitute 0 V for $v_1(0)$.

$$0 = A_1 e^{(0)} + A_2 e^{(0)} + 35$$

$$0 = A_1(1) + A_2(1) + 35$$

$$0 = A_1 + A_2 + 35$$

$$A_1 = -A_2 - 35 \dots\dots (3)$$

Step 13 of 17

Recall equation (2).

$$v_1(t) = A_1 e^{-t} + A_2 e^{-6t} + 35$$

Apply derivative with respect to t .

$$\begin{aligned} \frac{dv_1(t)}{dt} &= A_1 \frac{d(e^{-t})}{dt} + A_2 \frac{d(e^{-6t})}{dt} + \frac{d(35)}{dt} \\ &= -A_1 e^{-t} - 6A_2 e^{-6t} + 0 \\ &= -A_1 e^{-t} - 6A_2 e^{-6t} \end{aligned}$$

Substitute 0 for t .

$$\begin{aligned} \frac{dv_1(0)}{dt} &= -A_1 e^{-(0)} - 6A_2 e^{-6(0)} \\ &= -A_1 e^{(0)} - 6A_2 e^{(0)} \\ &= -A_1(1) - 6A_2(1) \\ &= -A_1 - 6A_2 \end{aligned}$$

Substitute 70 V/s for $\frac{dv_1(0)}{dt}$ and $-A_2 - 35$ for A_1 .

$$70 = -(-A_2 - 35) - 6A_2$$

$$70 = -5A_2 + 35$$

$$35 = -5A_2$$

$$A_2 = -7$$

Therefore, the value of the constant A_2 is -7 .

Step 14 of 17

Recall equation (3).

$$A_1 = -A_2 - 35$$

Substitute -7 for A_2 .

$$\begin{aligned} A_1 &= -(-7) - 35 \\ &= -28 \end{aligned}$$

Therefore, the value of the constant A_1 is -28 .

Step 15 of 17

Recall equation (2).

$$v_1(t) = A_1 e^{-t} + A_2 e^{-6t} + 35$$

Substitute -28 for A_1 and -7 for A_2 .

$$v_1(t) = 35 - 28e^{-t} - 7e^{-6t}$$

Therefore, the value of the voltage $v_1(t)$ is $35 - 28e^{-t} - 7e^{-6t}$.

Step 16 of 17

Recall equation (1).

$$\left(\frac{1}{2}\right) \frac{dv_1}{dt} = -2v_1 + v_2 + 35$$

$$\frac{dv_1}{dt} = -4v_1 + 2v_2 + 70$$

Substitute $35 - 28e^{-t} - 7e^{-6t}$ for $v_1(t)$.

$$\frac{d}{dt} \{35 - 28e^{-t} - 7e^{-6t}\} = -4(35 - 28e^{-t} - 7e^{-6t}) + 2v_2 + 70$$

$$\frac{d(35)}{dt} - 7 \frac{d(e^{-6t})}{dt} - 28 \frac{d(e^{-1t})}{dt} = -4(35 - 7e^{-6t} - 28e^{-1t}) + 2v_2 + 70$$

$$0 + 42e^{-6t} + 28e^{-1t} = -140 + 28e^{-6t} + 112e^{-1t} + 2v_2 + 70$$

$$-140 + 28e^{-6t} + 112e^{-1t} + 2v_2 + 70 - 42e^{-6t} - 28e^{-1t} = 0$$

$$-70 - 14e^{-6t} + 84e^{-1t} + 2v_2 = 0$$

$$-35 - 7e^{-6t} + 42e^{-1t} + v_2 = 0$$

$$v_2 = 35 + 7e^{-6t} - 42e^{-1t}$$

Therefore, the value of the voltage v_2 is $35 + 7e^{-6t} - 42e^{-1t}$.

Step 17 of 17

Calculate the value of the voltage $v_o(t)$.

$$v_o(t) = v_1(t) - v_2(t)$$

Substitute $35 + 7e^{-6t} - 42e^{-t}$ for v_2 and $35 - 28e^{-t} - 7e^{-6t}$ for $v_1(t)$.

$$\begin{aligned} v_o(t) &= 35 - 28e^{-t} - 7e^{-6t} - (35 + 7e^{-6t} - 42e^{-t}) \\ &= 35 - 28e^{-t} - 7e^{-6t} - 35 - 7e^{-6t} + 42e^{-t} \\ &= -14e^{-6t} + 14e^{-t} \\ &= 14(e^{-t} - e^{-6t}) \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v_o(t)$ is $14(e^{-t} - e^{-6t}) \text{ V}, t > 0$.